



# Little Things That Matter In The World Of Designing Tiny Connectors

There are several important aspects concerning connectors: their physical design, communication protocols and standards, and their connectivity concerning PCB versus the external world, both from the consumer and non-consumer perspectives. Wireless connectivity is also an emerging trend that is gradually revolutionising the way we communicate. Here is a discussion between some experts on the subject

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Connectors have come a long way since they were conceptualised and invented. With changing times, they have undergone many different modifications. And with the ongoing technological developments, they will certainly go through more.

Compared to several years ago, connectors nowadays have become high-end and miniaturised. Also, earlier we used to rely on European connector manufacturers for our needs, but with the entry of many Chinese and Indian connector manufacturers, the same quality of products is now available at a much lower cost.

Due to globalisation and increasing awareness concerning certification, certain standards have been now introduced for

India (as compared to European ones like CE mark, UL certification, etc, which we used to follow earlier). The flame-retardability characteristics of the connectors are also improving.

“Connectors are the heart of everything. Even if you design electronics with the best microcontroller, best semiconductor chips and best software/firmware, the failure of the connector means the failure of the product. If a connector fails, then the design fails. Therefore, connector selection is very vital,” says Mahendra Karandikar, GM-Developments & Projects, JISL.

Adding to that, Abhinav Behl, R&D engineer at Logic Fruit Technologies says, “Connectors help establish contact with an electronic device. While choosing a connector type, the first thing to consider should be its durability. Any misalignment should not cause failure in the connector or the PCB.”

## Considerations for selecting a connector

First go through the datasheet and specification of a particular connector to have clarity regarding details such as its current carrying capacity, voltage rating, and type of insulation material used. Then check the other aspects. For

